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School of Mathematical Sciences

Department of Computer Science

MSc Computer Science : Batch 2023-25, Semester I, Midterm

DA104: Probability and Stochastic processes

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Student Name (in block letters):

Date: 22 September 2023

Student Roll No:

Max Marks: 60

Signature:

Time: 2hrs

Answer the following questions. Each answer should contain a short explanation

1. State three axioms of probability. Prove that if $E \subset F$ then $\mathbb{P}(E) \leq \mathbb{P}(F)$ where E, F are two events. [3+2]
2. Let A and B be two events such that $\mathbb{P}(A) = \frac{3}{4}$ and $\mathbb{P}(B) = \frac{1}{3}$. Show that $\frac{1}{12} \leq \mathbb{P}(A \cap B) \leq \frac{1}{3}$. [3]
3. Answer the following: [3 × 5]
 - a) Prove that: $\mathbb{P}(A \cup B \cup C) = 1 - \mathbb{P}(A^c|B^cC^c)\mathbb{P}(B^c|C^c)\mathbb{P}(C^c)$ where A^c, B^c, C^c are complement of the events A, B, C respectively.
 - b) Three unbiased die are rolled. Let E be the event that the sum of the outcome of three die is even and E_i denote the event that the i^{th} die shows even. Are $E_i|E$ independent?
 - c) Let $A_r, r \geq 1$ be a sequence of events. Let $\mathbb{P}(A_r) = 1, r \geq 1$. What is $\mathbb{P}(A_1 \cap A_2)$? What is $\mathbb{P}(\bigcap_{i=1}^n A_i)$? Extend your arguments to compute $\mathbb{P}(\bigcap_{i=1}^{\infty} A_i)$.
 - d) A random number N of dice is thrown. Let A_i be the event that $N = i$. $\mathbb{P}(A_i) = 2^{-i}$ and let S denote the sum of the outcome. Compute the probability that $N = 2$ given $S = 4$.
 - e) A coin shows heads with probability p . What is the probability it will show three heads before five tails if the coin is tossed repeatedly? Assume tosses are independent.
4. An urn contains b blue balls and r red balls. The balls are removed at random and not replaced. Describe the sample space and show that each outcome is equally likely. Then show that the probability that the first red ball is the $(k+1)^{th}$ ball is $\frac{\binom{r+b-k-1}{r-1}}{\binom{r+b}{r}}$. [5]
5. Let X be a random variable with distribution function F . Then answer the following: [3 × 4]
 - a) What is the distribution function of $Y = aX + b$, where a, b are real constants?
 - b) Determine the distribution function of X^2 .
 - c) Determine the distribution function of $F(X)$. Assume F to be strictly increasing.
 - d) Let $X^+ = \max\{0, X\}$. Determine distribution function of X .
6. Suppose the distribution function of a random variable X is given by : [6]

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \leq x < 1 \\ \frac{x+1}{4} & 1 \leq x < 3 \\ 1 & 3 \leq x \end{cases}$$

Compute

- a) $\mathbb{P}(X < 1)$
- b) $\mathbb{P}(X > 2)$
- c) $\mathbb{P}(\frac{1}{3} < X < \frac{5}{3})$

7. State the probability mass function of a binomial distribution. Compute its mean and variance. [2+4+3]

8. Suppose X is a random variable such that $\mathbb{E}(X) = 3$ and $Var(X) = 1$. Compute the following: [5+5]

- a) $\mathbb{E}[(4X - 1)^2]$
- b) $Var(5 - 2X)$